

Passivity Preservation for Variable Impedance Control of Compliant Robots

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Abstract—The notable performance exhibited by impedance controllers during robotic interaction has led to the widespread use of this control methodology. Improved position and interaction control might be attainable through utilisation of variable impedance control (VIC) techniques. Interactional performance could be further improved by combining structural compliance with VIC. However, utilization of VIC tends to induce energy-injecting elements, which could impact on a robotic system's stability/passivity. Additionally, implementation of active VIC techniques on passively compliant robots has not been investigated (although several works consider VIC using variable stiffness actuators), which renders the existing rigid-joint robot, passivity-inducing control schemes inapplicable to compliant systems. To this end, the work presented here introduces a novel scheme, termed passivity-preservation control (PPC), which suppresses the energy injections that could be introduced into compliant robots, as a result of VIC. Compared to tank-based VIC approaches the PPC scheme is directly applicable to flexible-joint robots, even ones with nonlinear passive stiffness elements, while its performance is independent of the tank-energy levels. Moreover, the proposed scheme permits stable VIC using full-state feedback, thereby enabling impedance modulations relating to both motor and link-side variables. Consequently, full-state feedback gains can be generated via linear quadratic regulator optimisation, thus enabling application of gain-scheduling techniques on flexible-joint robots for enhanced position control. Passivity and stability analyses are performed for joint and Cartesian-space versions of the PPC scheme, which justify their applicability to both interaction and “free-motion” scenarios. The PPC scheme's efficacy, compared to constant gain impedance methods, in terms of convergence and interactional performance,

is corroborated via simulation and experimental means involving the Sawyer robot, which is powered by series elastic actuators. Theoretical and experimental results mathematically and practically verify VIC stability, thus enabling flexible-joint robots to more accurately mimic biologically inspired behaviors.

Index Terms—Human-robot interaction, impedance control, manipulators.

I. INTRODUCTION

THE increasing necessity for operational safety, as regards the interaction of robotic devices with their immediate environments, has stimulated collective research effort toward the resolution of the robotic interaction problem. Traditional robotic manipulators possessing “stiff” mechanical structures, typically operated in position-control mode, are deemed unsuitable for interaction tasks. Moreover, direct force control methods may exhibit insufficient levels of robustness when limited knowledge of the desired contact forces is available, or when environmental parameters are continuously fluctuating. Thus, the majority of environmental interaction control methods relies upon variants of impedance control, i.e., compliance control [5], stiffness control [6], and admittance control [6]. The impedance control concept was introduced to the field via [7], as a means of allowing robots to establish user-defined dynamical relationships between their motions and the associated external forces [8]. Robotics literature contains an array of works involving impedance controllers, including some of the most fundamental research outputs [7], [8], as well as additional examples of robustly performing impedance controllers [9], [10].

Supplementary interaction performance enhancements can be acquired by means of utilizing variable impedance control (VIC) instead of invariant-impedance control (IIC), as the former bears a closer resemblance to the operation of biological machines. Those biological machines display superior environmental adaptability in comparison to their artificial counterparts, as a consequence of their inherent impedance modulation mechanisms, hypothesised by [4]. The same work [4] proposes a reinforcement learning algorithm to select appropriate values for an impedance-gain scheduler. Ikeura and Inooka [11] describe a VIC method relying upon translation of human muscle activity to an array of active impedance gains for a robot controller. Tsumugiwa *et al.* [12] report on a technique that facilitates human–robot interaction between a manipulator and human operator, via rapid impedance modulations. Blaya and Herr [13] describe a biologically inspired study involving real-time

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VIC of an orthotic joint. For the purpose of permitting smooth adaptation to external impacts during human–robot interaction tasks, Yang *et al.* [14] propose a learning controller that gradually adjusts the robot’s total impedance and feedforward force signals. Calinon *et al.* [15] introduce a methodology that exploits a task-parameterized model, which enables the generation of optimal active stiffness and damping gains using linear quadratic regulator (LQR) optimization, in accordance with the poses of certain environmental artefacts. Similarly, Rozo *et al.* [16] extend the latter work by encoding additional state information, while devising an optimal control method to generate Cartesian impedance gain matrices. Gribovskaya *et al.* [17] describe an algorithm that combines adaptive control and programming by demonstration to regulate a robot’s impedance parameters. Moreover, Spyrakos-Papastavridis *et al.* [18] and [19] demonstrate that a humanoid robot can switch between heightened tracking accuracy and compliant interaction, using VIC, for the purpose of improving bipedal balancing performance. Ficuciello *et al.* [20] present a framework for the merging of VIC with redundancy resolution methodologies.

It is worth noting, however, that VIC is also achievable “mechanically,” by means of utilizing variable stiffness actuators (VSAs) [21], [22], as has been demonstrated on DLR’s David robot [23], [24]. Braun *et al.* [25] describe an algorithm that is capable of simultaneously designing motion and impedance trajectories for systems actuated by VSAs, thereby exhibiting improved performance over their fixed-gain counterparts. Compared to other compliant robot designs, VSAs display improved impact robustness and energy efficiency, extended dynamic capability, and enhanced task adaptability [24], albeit at the expense of higher dimensional requirements, in addition to increased design and control complexity.

A different mechanical means of augmenting a robot’s interactional performance is the incorporation of passive elastic elements into its physical structure [26], [27], [28]. An established, compliant-joint robot design approach entails the usage of series elastic actuators (SEAs) [29], [30], whose inherent flexibility renders the pertinent robotic systems compliant during interaction with their environments, thereby reducing the forces arising during the post-impact phase. However, this beneficial interaction feature introduces under actuation, reduces the plant’s natural frequency, and induces oscillatory dynamics, thereby amplifying the complexity of the position regulation/tracking problem [31]. Moreover, SEA-based robots are incapable of regulating their passive compliance in real-time, although their mechanical designs are typically more compact than those of VSAs, as they comprise less actuators. An established strategy entails the implementation of impedance controllers onto SEA-driven robots to harness their respective interactional attributes [31]. Although initial works introduced stabilizing control schemes for flexible-joint robots [32], [33], those methods were subsequently augmented [31] [34] via incorporation of motor inertia scaling terms, torque and torque derivative feedback [34], [35], and Cartesian impedance gains, in addition to passivity analyses.

Since impedance controllers are typically used for interaction tasks, it is frequently deemed appropriate to assess their

interaction capabilities by means of passivity analyses [35], [36], which rely upon external-wrench-based conditions.

Implementation of active VIC schemes on SEA-powered robots is reported in [18] and [19], from which this work distinguishes itself, as it proposes a novel analytical controller for stable VIC of compliant robots. Furthermore, the passivity-preservation control (PPC) scheme uses full-state feedback [35], [37], thereby enhancing robustness and permitting systematic gain generation.

In spite of the heightened interaction capabilities offered by VIC, this technique may endanger a system’s stability/passivity, as it tends to induce energy-injecting terms [1]. To this end, Ferraguti *et al.* [1] propose a tank-based approach to suppress the relevant energy injections and, thus, preserve stability/passivity. Ferraguti *et al.* [2] offer an extension to [1] by considering a time-varying admittance controller, whose passivity is analyzed using port-Hamiltonian representations. In contrast, Kronander and Billard [3] resolves the VIC-related energy-injection problem through a modified Lyapunov function, which imposes restrictions on the active impedance fluctuations.

The work described herein assumes the pre-existence of a continuous variable impedance trajectory (or strategy) and proposes a novel control scheme to guarantee its stable replication by a compliant-joint robot. Thus, the proposed scheme enables implementation of gain-scheduling methods [38] on flexible-joint robots, a topic that is briefly explored in this work, by means of linearizing a compliant manipulator’s dynamics around two points and using LQR techniques to generate a set of gains that the system can transition between. A similar gain-tuning approach is described in [39]. The proposed method’s contributions are synthesised as follows:

- 1) A stable VIC scheme for flexible-joint robots, considering time-varying active and passive stiffnesses.
- 2) Utilization of link position and velocity feedback gains.

The rest of this article is organized as follows. Section II provides the compliant joint model and problem statement, while Section III proposes the novel PPC scheme in conjunction with stability/passivity analyses. Section IV describes the impedance modulation and gain tuning techniques, Section V presents simulations and experimental results acquired during Sawyer’s execution of various interaction tasks. Finally, Section VI concludes this article.

II. IMPEDANCE CONTROL OF COMPLIANT-JOINT ROBOTS

A. Compliant Robot Model

An n -link flexible-joint robot model may be developed by following the approach described in [40], which results in

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{N}\dot{\mathbf{q}} + \mathbf{K}(\mathbf{q} - \boldsymbol{\theta}) + \boldsymbol{\tau}_g(\mathbf{q}) = \boldsymbol{\tau}_{ex} \quad (1)$$

$$\mathbf{J}\ddot{\boldsymbol{\theta}} + \mathbf{D}\dot{\boldsymbol{\theta}} - \mathbf{K}(\mathbf{q} - \boldsymbol{\theta}) = \boldsymbol{\tau}_m \quad (2)$$

where $\mathbf{q} \in \mathbb{R}^n$ and $\boldsymbol{\theta} \in \mathbb{R}^n$ denote the link and motor positions; $\mathbf{M} = \mathbf{M}(\mathbf{q}) \in \mathbb{R}^{n \times n}$, $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$, and $\mathbf{N} \in \mathbb{R}^{n \times n}$ represent the link inertia, Coriolis, and viscous damping matrices, respectively; $\boldsymbol{\tau}_g(\mathbf{q}) \in \mathbb{R}^n$ signifies the gravitational torque vector, while $\boldsymbol{\tau}_{ex} \in \mathbb{R}^n$ and $\boldsymbol{\tau}_m \in \mathbb{R}^n$ symbolize the external torque

and actuation torque vectors, respectively. Moreover, $\mathbf{K} \in \mathbb{R}^{n \times n}$ represents the passive spring stiffness matrix, while $\mathbf{J}, \mathbf{D} \in \mathbb{R}^{n \times n}$ are the motor inertia and damping matrices, respectively. It is noteworthy that

$$\|\tau_\partial\| = \|\partial\tau_g(q)/\partial q\| = \|\partial^2 U_g(q)/\partial q^2\| \leq \alpha \quad (3)$$

where $\alpha > 0$, and $U_g(q) = U_{g_q}$ signifies the gravitational potential, and $\tau_g(q) = (\partial U_g(q)/\partial q)^T$. The operator norm has been considered, i.e., $\|\mathbf{A}\| = \max\{\|\mathbf{A}x\|/\|x\|\}$.

B. Problem Formulation

A widely used regulation controller aimed at stable operation of compliant-joint robots is the following [32]:

$$\tau_m = \mathbf{K}_{P_M} \theta_E - \mathbf{K}_{D_M} \dot{\theta} + \tau_{gd} \quad (4)$$

wherein $\mathbf{K}_{P_M}$ and $\mathbf{K}_{D_M} \in \mathbb{R}^{n \times n}$ signify the symmetric, positive definite active stiffness and damping matrices, respectively, $\theta_E = \theta_d - \theta$, with $q_d, \theta_d \in \mathbb{R}^n$ being the referential link and motor positions, while $\tau_{gd} = \tau_g(q_d) \in \mathbb{R}^n$ encompasses the desired gravity compensation input signal's components.

Considering the system's steady-state form, which is acquired by setting $\dot{q}_F = \ddot{q}_F = 0$, with $q_F \in \mathbb{R}^{2n} = [q^T \ \theta^T]^T$ denoting the combined link and motor position vector, the overall stiffness matrix may then be represented as follows:

$$\mathbf{T}_I \in \mathbb{R}^{2n \times 2n} = \begin{bmatrix} \mathbf{K} & -\mathbf{K} \\ -\mathbf{K} & \mathbf{K} + \mathbf{K}_{P_M} \end{bmatrix}. \quad (5)$$

Combining the above with (3), leads to the condition [32]

$$\lambda_m(\mathbf{T}_I) \geq \alpha \quad (6)$$

wherein $\lambda_m(\cdot)$ and $\lambda_M(\cdot)$ represent the minimum and maximum eigenvalue of a given matrix. If $\mathbf{K}_{P_M}$ remains constant during operation (IIC), and $\lambda_m(\mathbf{T}_I) \geq \alpha$, then it is provable that there exists a unique stable equilibrium [32], [41] at $q_{eq} = [q_d^T \ \theta_d^T \ 0 \ 0]^T$, by considering the function [32]

$$V(q_F, \dot{q}_F) = \frac{\dot{q}_F^T \mathbf{M}_T \dot{q}_F}{2} + \frac{q_e^T \mathbf{T}_I q_e}{2} + U_{g_q} - U_{g_{q_d}} + q_{F_E}^T \begin{bmatrix} \tau_{gd} \\ 0 \end{bmatrix} \quad (7)$$

with $q_e \in \mathbb{R}^{2n} = q_F - q_{F_d}$, $q_{F_E} = q_{F_d} - q_F$, $q_{F_d} = [q_d^T \ \theta_d^T]^T$, where the overall inertia matrix $\mathbf{M}_T \in \mathbb{R}^{2n \times 2n} = \text{diag}([\mathbf{M} \ \mathbf{J}])$. Calculation of the above function's time derivative, accompanied by several simplifications, produces

$$\dot{V}(\dot{q}_F) = -\dot{q}_F^T \Psi \dot{q}_F \quad (8)$$

where $\Psi \in \mathbb{R}^{2n \times 2n} = \text{diag}([0 \ \mathbf{D} + \mathbf{K}_{D_M}])$ (neglecting $\mathbf{N}$), thus permitting a proof of global asymptotical stability (GAS) when applying LaSalle's Principle (LP). One can also establish the closed-loop system's passivity during interaction ($\tau_{ex} \neq 0$) by exploiting the passivity-related lemmas contained in [36], yielding the relationship

$$\dot{Q} = \dot{q}^T \tau_{ex} - \dot{q}_F^T \Psi \dot{q}_F \leq \dot{q}^T \tau_{ex} \quad (9)$$

where Q signifies a storage function. Satisfaction of the passivity condition is guaranteed when Ψ is positive definite.

The active impedances may be varied in real time by modulating the proportional-derivative (PD) gains using

$$\tau_m = \mathbf{K}_{P_{Mt}} \theta_E - \mathbf{K}_{D_{Mt}} \dot{\theta} + \tau_{gd} \quad (10)$$

where $\mathbf{K}_{P_{Mt}} = \mathbf{K}_{P_M}(t)$ and $\mathbf{K}_{D_{Mt}} = \mathbf{K}_{D_M}(t)$. Computation of (7)'s time derivative produces a modified formula w.r.t (8), as demonstrated through the following equation:

$$\dot{V}(\dot{q}_F, q_e) = -\dot{q}_F^T \Psi \dot{q}_F + \frac{1}{2} q_e^T \dot{\mathbf{T}}_I q_e \quad (11)$$

where $\dot{\mathbf{T}}_I \in \mathbb{R}^{2n \times 2n} = \text{diag}([0 \ \dot{\mathbf{K}}_{P_{Mt}}])$, since $\mathbf{K}$ is a time-invariant matrix. The indeterminate nature of the right-most stiffness-related element may destroy the closed-loop system's stability for increasing $\mathbf{K}_{P_{Mt}}$ values. Conversely, active stiffness decreases ensure stability preservation.

C. Augmented-State Controllers

The vast majority of state-of-the-art PD-plus-type controllers for flexible-joint robots relies upon the exploitation of indirect forms of link position/velocity feedback [31], [34], [35], coupled with inner torque loops. This feature augments the controllers' dimension and enhances robustness, by accounting for the link (in addition to the motor) states, which are directly interacting with the environment. Even though [32] accounts solely for motor-side feedback, the robot considered in the relevant simulations [32] possesses passive stiffness values exceeding those of some contemporary SEA-powered robots [24], [27] by more than an order of magnitude.

III. PASSIVITY-PRESERVATION CONTROL

A. Flexible-Joint Robot Variable Impedance Controller

Stable VIC of flexible-joint robots relies upon removal of the adverse stability-related effects associated with the energy-injecting element, which is achievable via the PPC law

$$\tau_m = \mathbf{K}_{P_{Mt}} \theta_E - \mathbf{K}_{D_{Mt}} \dot{\theta} + \mathbf{K}_{P_{Jt}} q_E - \mathbf{K}_{D_{Jt}} \dot{q} + \tau_{gd} - \frac{1}{2} \dot{\Theta}_{PI} I_d \left(q_e^T \dot{\mathbf{T}}_K q_e + \dot{q}_F^T (\mathbf{T}_K^T - \mathbf{T}_K) q_e \right) \quad (12)$$

where $q_E = q_d - q$, with $\dot{\Theta}_{PI} \in \mathbb{R}^{n \times n} = [\text{diag}(\dot{\theta})]^+$ signifying the Moore–Penrose pseudoinverse of the diagonalized motor velocity vector $\dot{\theta}$. Moreover, $I_d = [1 \dots 1_n]^T \cdot \text{rank}(\dot{\Theta}_{PI})^+ \in \mathbb{R}^n$ symbolizes a 'rank-normalized' unit vector, $\mathbf{K}_{P_{Jt}}$ and $\mathbf{K}_{D_{Jt}} \in \mathbb{R}^{n \times n}$ denote time-varying, link-side PD gains, respectively, while

$$\mathbf{T}_K = \begin{bmatrix} \mathbf{K} & -\mathbf{K} \\ -\mathbf{K} + \mathbf{K}_{P_{Jt}} & \mathbf{K} + \mathbf{K}_{P_{Mt}} \end{bmatrix}. \quad (13)$$

Since controller (12) considers both motor and link feedback, it aligns more closely with state-of-the-art schemes [31], [34], [35], and it essentially functions by modifying the damping levels associated with joints that are either experiencing stiffness variations or possessing $\mathbf{K}_{P_{Jt}}$ gains.

Theorem 1: If the closed-loop system comprising (1), (2), and (12), satisfies $\lambda_m(\mathbf{T}_K) > \alpha$ and $\lambda_m(\Xi_D) > 0$, and T_K is positive definite, then this ensures the existence of a GAS equilibrium at q_{eq} .

Proof 1: Proving asymptotical stability of the time-varying closed-loop system necessitates the application of Barbalat's lemma, followed by LP. Thus, it is crucial to demonstrate the uniqueness of the equilibria (see Appendix A), such that $V = 0$ only when $q_{F_E} = \dot{q}_F = 0$, if the following holds:

$$\lambda_m(\mathbf{T}_K) > \alpha. \quad (14)$$

The following potential energy function is, thus, utilized [32]:

$$W = \frac{1}{2} q_e^T \mathbf{T}_K q_e + U_g(q) - U_g(q_d) - q_e^T \begin{bmatrix} \tau_{gd} \\ 0 \end{bmatrix} \quad (15)$$

whose unique equilibrium point is proven via the process outlined in Appendix A. The total Lyapunov function is thus

$$V = V(q_F, \dot{q}_F) = \frac{1}{2} \dot{q}_F^T \mathbf{M}_T \dot{q}_F + W. \quad (16)$$

Obtaining V 's time derivative (see Appendix B), yields

$$\dot{V} = \dot{V}(\dot{q}_F) = -\dot{q}_F^T \Psi_K \dot{q}_F \quad (17)$$

where the total damping matrix possesses the form

$$\Psi_K \in \mathbb{R}^{2n \times 2n} = \begin{bmatrix} \mathbf{N} & 0 \\ \mathbf{K}_{D_{Jt}} & \mathbf{D} + \mathbf{K}_{D_{Mt}} \end{bmatrix}. \quad (18)$$

Preservation of $\dot{V}$'s negative definiteness requires that Ψ_K be positive definite, which is guaranteed when the condition

$$\Xi_D = \mathbf{N} - \frac{1}{4} \mathbf{K}_{D_{Jt}}^T (\mathbf{D} + \mathbf{K}_{D_{Mt}})^{-1} \mathbf{K}_{D_{Jt}} > 0 \quad (19)$$

is satisfied (see Appendix B). However, convergence is only ensured when $\lim_{t \rightarrow \infty} \dot{V}(\dot{q}_F, t) = 0$, and since the system is time-varying (owing to VIC), the time derivative of $\dot{V}$ is

$$\ddot{V} = \ddot{V}(\dot{q}_F, \ddot{q}_F) = -\ddot{q}_F^T (\Psi_K + \dot{\Psi}_K) \dot{q}_F - \dot{q}_F^T \dot{\Psi}_K \dot{q}_F. \quad (20)$$

Application of Barbalat's lemma, together with an extended form of LP, proves GAS of the equilibria (see Appendix B).

B. Passivity-Preservation Analysis

The effect of external perturbations/contacts on the system's performance may be assessed through a passivity analysis.

Theorem 2: If system (1), (2), and (12) fulfils $\lambda_m(\Xi_D) > 0$, then this ensures the production of a passive system.

Proof 2: Constructing a closed-loop system comprising (1), (2), and (12), while using the storage function (7) and assuming that $\tau_{ex} \neq 0$, it is demonstrable that (see Appendix B)

$$\dot{Q} = \dot{q}^T \tau_{ex} - \dot{q}_F^T \Psi_K \dot{q}_F \leq \dot{q}^T \tau_{ex} \quad (21)$$

where Q denotes a storage function. Positive definiteness of Ψ_K is a prerequisite for the validity of (21). Thus, the user is allowed to modulate the system's impedances in real-time to achieve the desired interaction performance, while ensuring passivity. Although similar general results exist [35], (21) proves VIC

passivity for the novel PPC scheme, containing the physically intuitive link position/velocity gains, while enabling consideration of nonlinear passive stiffnesses.

C. Variable Cartesian Impedance Controller

A Cartesian version of the previously described joint-space controller, leading to the creation of a variable Cartesian impedance control (VCIC) scheme, is shown as

$$\begin{aligned} \tau_m = & \mathbf{J}_t^T (\mathbf{K}_{ct} x_E - \mathbf{D}_{ct} \dot{x}) + \mathbf{K}_{P_{Jt}} q_E - \mathbf{K}_{D_{Jt}} \dot{q} \\ & + \tau_{gd} - \frac{1}{2} \dot{\Theta}_{PI} I_d \left(x_{ie}^T \dot{\mathbf{K}}_{ct} x_{ie} + q_e^T \dot{\mathbf{T}}_{KJ} q_e \right. \\ & \left. + \dot{q}_F^T (\mathbf{T}_{KJ}^T - \mathbf{T}_{KJ}) q_e \right) \end{aligned} \quad (22)$$

where $\mathbf{T}_{KJ} = \begin{bmatrix} \mathbf{K} & -\mathbf{K} \\ -\mathbf{K} + \mathbf{K}_{P_{Jt}} & \mathbf{K} \end{bmatrix}$, $\mathbf{J}_t^T \in \mathbb{R}^{n \times m}$ denotes a Jacobian matrix performing the joint-to-Cartesian space mapping, as per $\dot{x} = \mathbf{J}_t \dot{\theta}$, with m signifying the task variables. Moreover, $\mathbf{K}_{ct}$ and $\mathbf{D}_{ct} \in \mathbb{R}^{m \times m}$ represent time-varying Cartesian stiffness and damping matrices, respectively, $x_E = x_d - x$ and $x_{ie} = x - x_d$. It is demonstrable that the system possesses a unique equilibrium, by considering the function

$$\begin{aligned} V = & \frac{1}{2} \dot{q}_F^T \mathbf{M} \dot{q}_F + \frac{1}{2} x_{ie}^T \mathbf{K}_{ct} x_{ie} + \frac{1}{2} q_e^T \mathbf{T}_{KJ} q_e \\ & + U_g(q) - U_g(q_d) + q_{F_E}^T \begin{bmatrix} \tau_{gd} \\ 0 \end{bmatrix} \end{aligned} \quad (23)$$

which is equatable with (7) that had been proven to possess a unique equilibrium. Hence, for the Cartesian case

$$\mathbf{T}_K = \mathbf{T}_C = \begin{bmatrix} \mathbf{K} & -\mathbf{K} \\ -\mathbf{K} + \mathbf{K}_{P_{Jt}} & \mathbf{K} + \mathbf{J}_t^T \mathbf{K}_{ct} \mathbf{J}_t \end{bmatrix} \quad (24)$$

since $\mathbf{K}_{P_{Mt}} = \mathbf{J}_t^T \mathbf{K}_{ct} \mathbf{J}_t$ (and $\mathbf{K}_{D_{Mt}} = \mathbf{J}_t^T \mathbf{D}_{ct} \mathbf{J}_t$). Following a series of calculations (see Appendix C), the resulting derivative of the previously presented energy function is the following:

$$\dot{V} = -\dot{q}_F^T \Psi_{K_C} \dot{q}_F \quad (25)$$

where $\Psi_{K_C} = \begin{bmatrix} \mathbf{N} & 0 \\ \mathbf{K}_{D_{Jt}} & \mathbf{D} + \mathbf{J}_t^T \mathbf{D}_{ct} \mathbf{J}_t \end{bmatrix}$. Equation (25) permits concluding GAS, by following the approach described in Appendix B.

D. Extension to Nonlinear Stiffness and Friction Models

For improved modelling accuracy, it is possible to utilize the following link-side dynamical model [together with (2)]:

$$\mathbf{M} \ddot{q} + (\mathbf{C} + \mathbf{N}) \dot{q} + \mathbf{K}(t)(q - \theta) + \tau_f + \tau_g(q) = \tau_{ex} \quad (26)$$

where $\mathbf{K}(t) \in \mathbb{R}^{n \times n}$ denotes a time-varying passive stiffness matrix, while $\tau_f \in \mathbb{R}^n$ signifies the frictional torque vector comprising Stribeck, Coulomb, and static friction terms. In order to preserve closed-loop stability, the following adapted PPC scheme could be used (for joint impedance control):

$$\tau_m = \mathbf{K}_{P_{Mt}} \theta_E - \mathbf{K}_{D_{Mt}} \dot{\theta} + \mathbf{K}_{P_{Jt}} q_E - \mathbf{K}_{D_{Jt}} \dot{q} + \tau_{gd}$$

$$-\frac{1}{2}\dot{\Theta}_{PI}I_d\left(q_e^T\dot{T}_Kq_e+\dot{q}_F^T(T_K^T-T_K)q_e-2\dot{q}^T\tau_f\right). \quad (27)$$

The effect of $K(t)$ is suppressed by virtue of $q_e^T\dot{T}_Kq_e$, whereas the frictional effects are compensated via the $\dot{q}^T\tau_f$ term. Controller (27) is introduced for the purpose of demonstrating extensibility to systems possessing disparate mechanical properties, w.r.t the Sawyer robot.

IV. ACTIVE IMPEDANCE GAIN GENERATION

A. Contact-Based Impedance Modulation

It has been suggested that low impedances may be exploited for external force mitigation during interaction [5], while high impedance values permit enhanced tracking/regulation accuracy in “free motion” [5]. Thus, the VIC approach should entail smooth, continuous impedance fluctuations determined by the values of the external contact wrenches, as is the paradigm for VIC [1], [3], [10], [12], [15]. In this manner, the robot will be capable of adapting its behavior in accordance with the environmental conditions (contacts), thereby rendering it compliant and safer during interaction—a feature that is precluded by IIC. Transitioning between the various impedance gain sets could, therefore, be performed by means of a ramp function, as per the law

$$K_{O_j} = \begin{cases} K_{O_{j-1}} + (K_H - K_{O_{j-1}})(j \cdot n_r^{-1}), & \text{if } c_1 \\ K_{O_{j-1}} - (K_{O_{j-1}} - K_L)(j \cdot n_r^{-1}), & \text{if } c_2 \end{cases} \quad (28)$$

where $c_1 = (F_{exN} < F_{exTh}) \wedge (\tau_{exN} < \tau_{exTh})$, $c_2 = (F_{exN} \geq F_{exTh}) \vee (\tau_{exN} \geq \tau_{exTh})$, j is the sample number, and n_r signifies the total number of ramping samples. Furthermore, $K_O \in \mathbb{R}^{n \times 4n}$, $K_H \in \mathbb{R}^{n \times 4n}$, $K_L \in \mathbb{R}^{n \times 4n}$, $F_{ex} \in \mathbb{R}^3$, and $\tau_{exw} \in \mathbb{R}^3$ represent the total gain matrix, minimum and maximum active impedance gain matrices, and external force and moment vector components of the wrench vector, respectively. $F_{exN} = \|F_{ex}\|$ and $\tau_{exN} = \|\tau_{exw}\|$, while F_{exTh} and τ_{exTh} symbolize external force and moment thresholds. Alternative gain transitioning methods are also available [3].

B. LQR-Based Impedance Gain Generation

Aside from their straightforward physical intuition and consideration of interaction port states, one of the main benefits of control laws (12) and (22) is that they facilitate the challenging task of full-state feedback gain generation, which would otherwise involve the manual tuning of four distinct gain matrices. Although sophisticated model-based, flexible-joint robot gain-tuning methods are available [42], they lead to the production of gain matrices tailored to control designs involving torque and torque derivative feedback [42]. Contrarily, LQR can be used for direct generation of diagonal PPC gains, given that numerous systematic LQR gain matrix diagonalization techniques are available. Furthermore, LQR enables minimization of both position and force error metrics [43], [44], and constitutes a formal, nonheuristic, model-based impedance-gain adaptation method [15], [16]. Since control laws (12) and (22) contain

$\tau_g(q_d)$, one should set $\tau_g(q) = 0$ in (1), prior to linearization [39], thereby yielding

$$\dot{x} = \begin{bmatrix} 0 & \mathbf{I} \\ -M_0^{-1}P_0 & -M_0^{-1}N_0 \end{bmatrix} x + \begin{bmatrix} 0 \\ M_0^{-1}\tau_{se} \end{bmatrix} \tau_m. \quad (29)$$

In the aforementioned equation, $M_0 \in \mathbb{R}^{2n \times 2n} = \text{diag}([M_L \ J])$, $N_0 \in \mathbb{R}^{2n \times 2n} = \text{diag}([C_0 + N \ D])$, and $P_0 \in \mathbb{R}^{2n \times 2n}$ denotes the passive elasticity matrix; M_L and $C_0 \in \mathbb{R}^{n \times n}$ signify the linearized link inertia and Coriolis matrices, respectively; $\tau_{se} \in \mathbb{R}^{2n \times n}$ symbolizes a selection matrix, and $x = [q^T \ \theta^T \ \dot{q}^T \ \dot{\theta}^T]^T$. The following representation is considered:

$$\dot{x} = Ax + Bu \quad (30)$$

$$y = Cx \quad (31)$$

where $C = [I \ 0]$, $u = \tau_m$; and $A \in \mathbb{R}^{4n \times 4n}$ and $B \in \mathbb{R}^{4n \times n}$ denote the state transition and input matrices, respectively. The LQR problem is solved by minimizing the cost function

$$S = \sum_{k=0}^{\infty} \{x^T Q x + \tau_m^T R \tau_m\} \quad (32)$$

subject to the system's dynamics, where $Q \in \mathbb{R}^{4n \times 4n}$ and $R \in \mathbb{R}^{n \times n}$ denote diagonal and positive definite penalty matrices. The latter yields a solution to the algebraic Riccati equation

$$A^T P_u + P_u A - P_u B R^{-1} B^T P_u + Q = 0 \quad (33)$$

where $P_u \in \mathbb{R}^{4n \times 4n}$ denotes the unknown matrix, whose value is utilized to compute the following gain matrix:

$$K_{LQR} \in \mathbb{R}^{n \times 4n} = R^{-1} B^T P_u \quad (34)$$

thus leading to the creation of the control law

$$u = -K_{LQR}x + G_{ff}y_{ref} \quad (35)$$

where $G_{ff} \in \mathbb{R}^{n \times n}$ signifies a feedforward matrix and $y_{ref} \in \mathbb{R}^n$ denotes the referential position vector.

C. Controller Augmentation and Gain Adaptation

To utilize diagonal gain matrices conforming to the stability conditions, the original controller design has been decentralized [39]. For each gain-set, closed-loop stability is assessed by determining that the linearized system satisfies $A_{eNG} = \lambda_M(A_{cl}) < 1$, where $A_{cl} = A - BK_{LQR}$. Moreover, the generated gains should satisfy $\lambda_m(T_K) > \alpha$, which is achieved by assigning larger values to the Q matrix penalty terms associated with motor position states, as opposed to link position states, such that $K_{P_{Mt}}$ becomes the dominant term in T_K , as $K_{D_{Mt}}$ should also be the dominant term in Ψ_K .

Superimposition of the gravity compensation and passivity-preservation signals, onto (35), leads to (12) and/or (22).

Low and high impedance designs, pertaining to the contact, and “free-motion” scenarios have been acquired by following [39]’s approach. Solving the time-varying LQR problem [45] together with [39] could enable the generation of contact-aware impedance gains for enhanced interaction performance.

V. VARIABLE IMPEDANCE INTERACTION EXPERIMENTS

The Cartesian-space controller described by (22) is hereafter referred to as VCIC1. To assess the PPC scheme's efficacy, in terms of interaction task execution, VCIC1 has been implemented onto the Rethink Robotics Sawyer robot.

The experimental setup encompasses an Intel Core i7-6820HQ @ 2.7-GHz laptop running Robot Operating System (ROS) Kinetic, through Ubuntu 16.04 LTS (Xenial).

A. Contact-Dependent Variable Cartesian Impedance

The robot has been subjected to a series of external perturbations, which trigger ramp-based transitions between the predefined low and high impedance gain-sets, in accordance with user-adjusted external wrench thresholds. Two additional controllers are considered henceforth, namely, IIC1 and VCIC2. IIC1 is based on the full-state feedback control scheme proposed in [35], which possesses the form

$$\tau_m = \mathbf{J}_{\bar{q}}^T (\mathbf{K}_c x_E - \mathbf{D}_c \dot{x}) - \mathbf{K}_T \tau - \mathbf{K}_S \dot{\tau} + \mathbf{J} \mathbf{J}_S^{-1} \tau_g(\bar{q}) \quad (36)$$

where $\mathbf{J}_{\bar{q}} = \mathbf{J}_{\bar{q}}(\bar{q})$, with $\bar{q}$ denoting a modified position variable [35], $\tau = \mathbf{K}(\theta - q)$, $\mathbf{J}_S \in \mathbb{R}^{n \times n}$ signifies an inertia scaling matrix, while $\mathbf{K}_T$, $\mathbf{K}_S \in \mathbb{R}^{n \times n}$ represent torque and torque derivative gain matrices. VCIC2 is a time-varying version of IIC1. The VCIC1 gains corresponding to the contact and free-motion cases, respectively, are tuned using

$$\mathbf{Q}_c = \text{diag}([3 \ 5 \ 9 \ 9 \ 5 \mathbf{I}_3 \ 2.3 \ 3.8 \ 1.6 \ 1.6 \ 36 \ 12 \ 1.3 \ 10^{-8} \mathbf{I}_{14}])$$

$$\mathbf{R}_c = \text{diag}(10^{-4} [8 \ 16 \ 12 \ 10 \ 14 \ 30 \ 9])$$

$$\mathbf{Q}_f = \text{diag}$$

$$([3.5 \ 7 \ 11 \ 12 \ 0.5 \mathbf{I}_3 \ 2.3 \ 3.8 \ 1.6 \mathbf{I}_{246 \ 32 \ 1 \ 10^{-8} \mathbf{I}_{14}}])$$

$$\mathbf{R}_f = \text{diag}(10^{-5} [7 \ 10.5 \ 13.5 \ 23 \ 7.5 \ 27.5 \ 5])$$

giving rise to the following values: $\lambda_m(\mathbf{T}_{\mathbf{K}_c}) = 143.44$, $\lambda_m(\mathbf{\Xi}_{\mathbf{D}_c}) = 0.58$, $\lambda_m(\mathbf{T}_{\mathbf{K}_f}) = 548.22$, and $\lambda_m(\mathbf{\Xi}_{\mathbf{D}_f}) = 0.08$. The c and f subscripts denote the contact and free-motion phases, respectively. It is assumed that $k_i = 1000$ Nm/rad ($\mathbf{K} = \text{diag}([k_i, \dots, k_n])$), i.e., an order of magnitude smaller than the physical value, and significantly larger than the calculated $\alpha = 101.36$ value, which corresponds to the minimum joint stiffness ensuring a viable mechanical design for a compliant manipulator [46]. The conservativeness of k_i guarantees that $\lambda_m(\mathbf{T}_{\mathbf{K}}) > \alpha$. IIC1 and VCIC2 utilize $\mathbf{K}_c$, $\mathbf{D}_c$ values identical to those employed by VCIC1, albeit with different $\mathbf{K}_T$, $\mathbf{K}_S$ matrices [35]. Fig. 1 reveals that $\dot{Q} \leq \dot{q}^T \tau_{ex}$ when using VCIC1, while the relationship $\dot{Q} > \dot{q}^T \tau_{ex}$, could occasionally arise when performing VIC using controller VCIC2, thus potentially impacting on the system's stability/passivity. Fig. 2 displays the associated active stiffness fluctuations.

B. Smoothing Task: Experimental Setup

To verify the system's ability to maintain contact with a surface for a finite duration, while performing a predefined

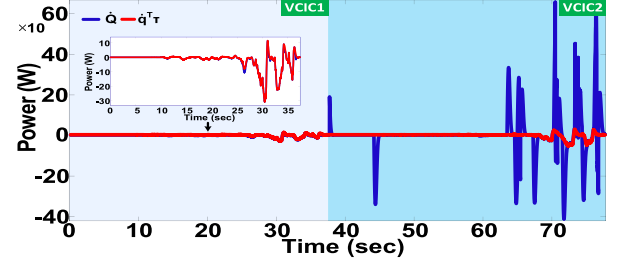


Fig. 1. Evolution of power variables (VCIC1 vs. VCIC2).

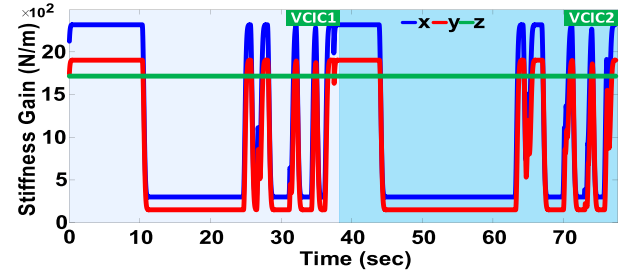


Fig. 2. Stiffness variations during manual disturbances (VCIC1 vs. VCIC2).

Cartesian motion, the robot has been commanded to establish contact with a surface located at a fixed distance, albeit possessing an angular offset w.r.t the Sawyer's base and vertical axis. To this end, the robot is provided with a set of Cartesian reference positions corresponding to a predetermined path mimicking a human-like task and is equipped with a plastering trowel (video file). In order to replicate the defined Cartesian path, which is updated via c_{td} , a Sawyer-specific ROS package [47] is used to compute the link-position Jacobian, and hence the desired joint velocities

$$\dot{q}_d = \mathbf{J}_q^+ \dot{c}_{td} = \mathbf{J}_q^+ \cdot \delta c_{td} \quad (37)$$

where $c_{td} \in \mathbb{R}^m$, $\mathbf{J}_q \in \mathbb{R}^{m \times n}$, and δ denote the referential Cartesian pose vector, link-position Jacobian, and delta operator, respectively. Hence, the referential Cartesian position increments are used to define linear task-space paths, corresponding to vertical smoothing and contact-plane motions. To suppress the dynamical production of desired positions, an array of incremental set points is generated by extracting the outputs of the above equation at less frequent, constant time intervals, as suggested by relevant methodologies [48]. Therefore, the desired positions are computed recursively via the relationship [49]

$$q_{d_i} = q_{d_{i-1}} + \dot{q}_{d_i} \Delta T \quad (38)$$

where $\Delta T = t_i - t_{i-1}$, which enables computation of the referential motor position terms present in the control law. Link to motor reference conversions are performed through

$$\theta_d = q_d + \mathbf{K}^{-1} \tau_{gd}. \quad (39)$$

In order to select nonsingular configurations for operation [4], [31], the robot is commanded to a predefined homing configuration (defined in terms of joint angles) at which the wrist joint occupies a fixed position away from the known structural singularities. This interpolation is performed using

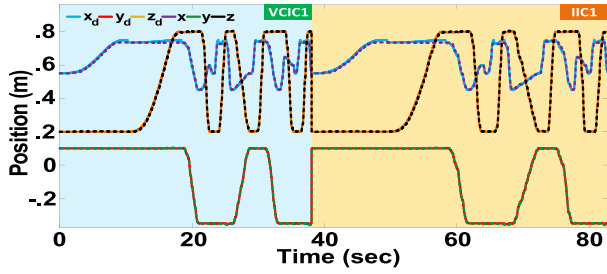


Fig. 3. Desired and actual Cartesian positions (VCIC1 vs. IIC1).

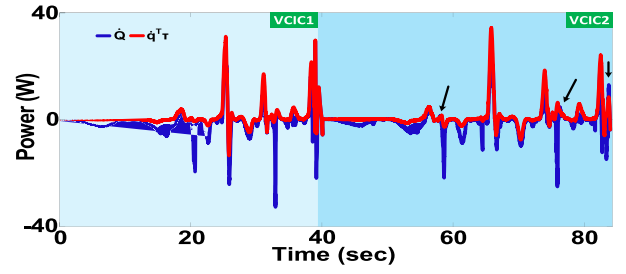


Fig. 6. Evolution of power variables (VCIC1 vs. VCIC2).

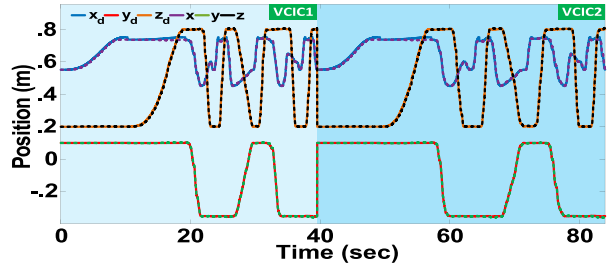


Fig. 4. Desired and actual Cartesian positions (VCIC1 vs. VCIC2).

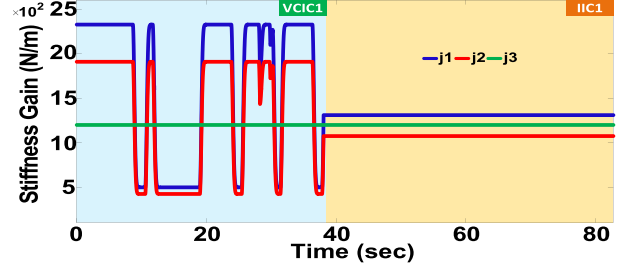


Fig. 7. Active Cartesian stiffness modulation (VCIC1 vs. IIC1).

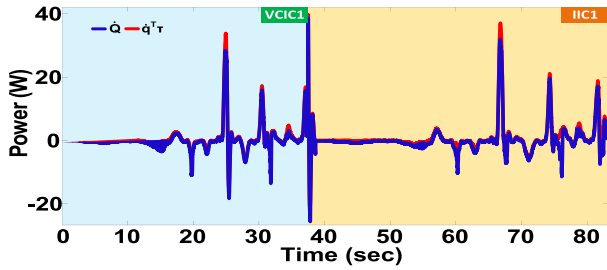


Fig. 5. Evolution of power variables (VCIC1 vs. IIC1).

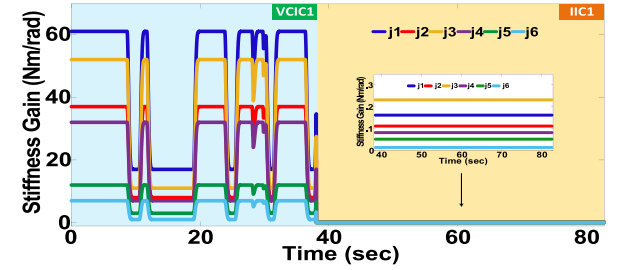


Fig. 8. Active link stiffness gain modulation (VCIC1 vs. IIC1).

the “move_to_neutral” method, contained in the Itera SDK ROS packages. Transitioning between the two distinct wrist configurations requires that the robot initially comes to rest at the final position of the Cartesian path segment, thereby satisfying the predefined Cartesian error thresholds, $|c_t - c_{td}| \leq c_{thr}$. Subsequently, execution of an existing joint interpolation algorithm initiates the desired wrist configuration switch. Contrarily, since the elbow and shoulder singularities are characterised in task-space [5], their avoidance necessitates operation away from the relevant Cartesian-space regions [5], i.e., regions that exceed the horizontal and vertical extremities of the task-space for a given contact-plane configuration.

C. Smoothing Task: VCIC Versus IIC

VCIC1 and VCIC2 modulate the robot’s impedances in accordance with (28). The displayed figures are divided into yellow (IIC) and blue (VCIC) subplots. Figs. 3 and 4 illustrate the actual and referential Cartesian positions measured during smoothing, while Figs. 5 and 6 depict the $\dot{Q}$ and $\dot{q}^T \tau_{ex}$ values associated with the relevant controllers.

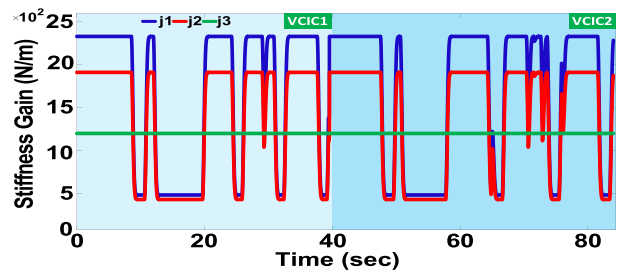


Fig. 9. Active Cartesian stiffness modulation (VCIC1 vs. VCIC2).

The latter results reveal that in contrast to VCIC1, utilization of VCIC2 could lead to the phenomenon $\dot{Q} > \dot{q}^T \tau_{ex}$. Furthermore, Figs. 7–10 display the fluctuating stiffness and damping gains; Figs. 11 and 12 depict the input torques, while Figs. 13 and 14 exhibit the link position values.

The external wrenches (smoothed using a 70-sample moving average filter to avoid unnecessary impedance switches) are displayed in Figs. 15 and 16. The results indicate that usage of IIC induces larger external forces during interaction as compared

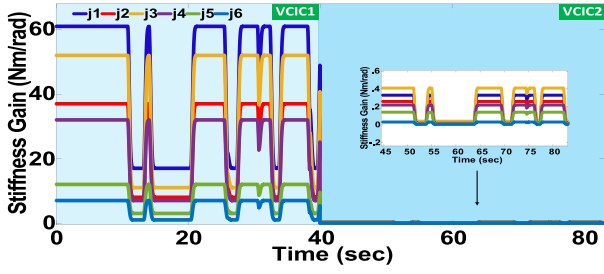


Fig. 10. Active link stiffness gain modulation (VCIC1 vs. VCIC2).

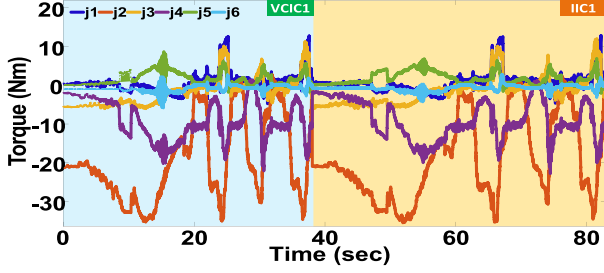


Fig. 11. Input torque signals (VCIC1 vs. IIC1).

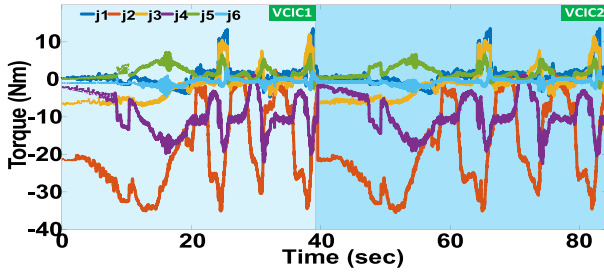


Fig. 12. Input torque signals (VCIC1 vs. VCIC2).

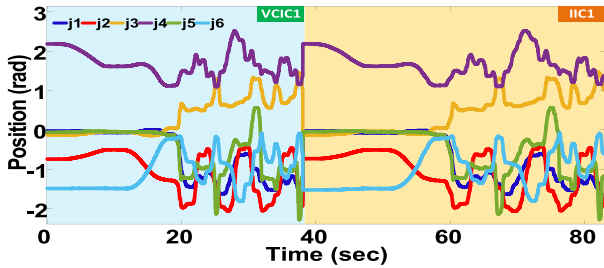


Fig. 13. Link position values (VCIC1 vs. IIC1).

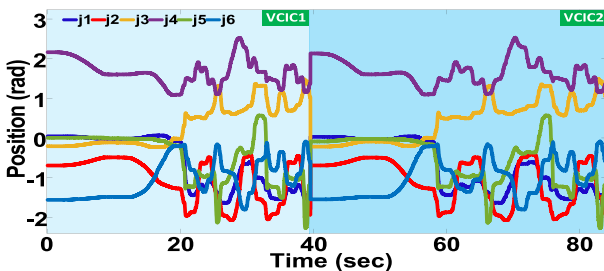


Fig. 14. Link position values (VCIC1 vs. VCIC2).

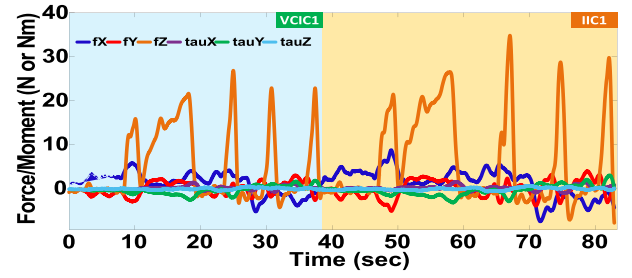


Fig. 15. Wrench variables (VCIC1 vs. IIC1).

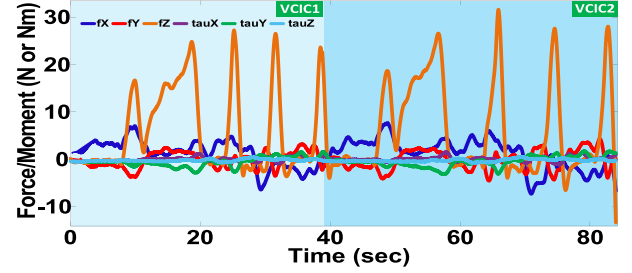


Fig. 16. Wrench variables (VCIC1 vs. VCIC2).

to VIC. On the other hand, VCIC2 displays lower and higher contact force magnitudes in comparison to IIC1 and VCIC1, respectively, while it is occasionally observed that $\dot{Q} > \dot{q}^T \tau_{ex}$ (indicated by arrows).

D. Practical Considerations

Despite the significant differences between PPC and classical PD-plus control, identical VIC strategies can be used with both methods in principal, although this depends on the noise levels, gain selection, and passive stiffness values. However, the stability analysis indicates that $\dot{q}_F = 0$ when $q_e = 0$, which naturally leads to minimization of the PPC scheme's auxiliary term magnitudes. For practical implementation, one should account for feedback noise, which may induce large input torque spikes at low velocities, owing to $\dot{\Theta}_{PI}$. Thus, impedance modulations could be prevented ($\dot{k}_{P_{Mti}} = \dot{k}_{KJ_i} = 0$) and $k_{P_{Jti}}$ ramped to zero, when $\|\dot{\theta}_i\| \leq n_{\dot{\theta}}$, with $n_{\dot{\theta}}$ denoting the signal noise magnitude and i indicating a matrix's i th diagonal element. Since the auxiliary elements are modifiable in real-time, the user is granted full control over the additive control signal. Moreover, PPC guarantees passivity during VIC, in contrast to classic PD-plus control.

E. Simulation-Based IIC Versus VCIC Comparison

A comparison between the IIC1 and VCIC1 schemes is performed through simulation of a 19 kg, three-DoF (similar to Sawyer's pitch joints), SEA-powered robotic manipulator, whose joint stiffnesses have been set to 3000 Nm/rad.

The simulation is performed by means of integrating the dynamical equations (1) and (2). For both the IIC and VCIC

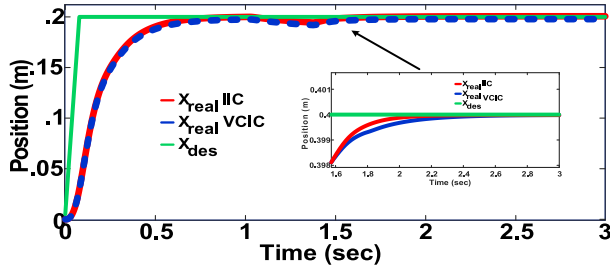


Fig. 17. Cartesian position reference and responses (VCIC1 vs. IIC1).

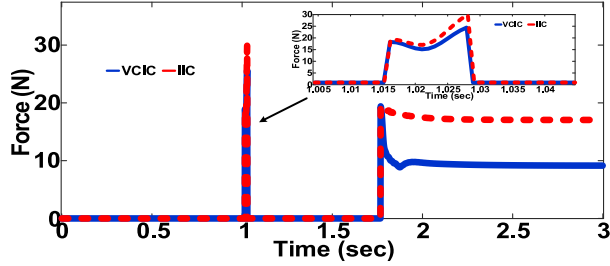


Fig. 18. Impact forces arising during contact (VCIC1 vs. IIC1).

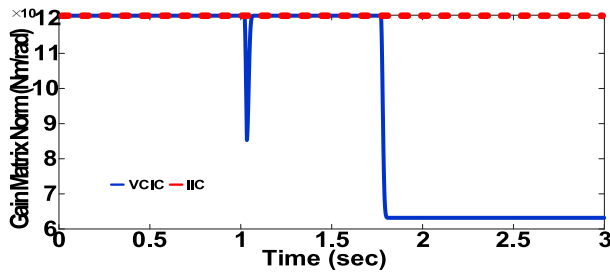


Fig. 19. Total stiffness gain matrix norm (VCIC1 vs. IIC1).

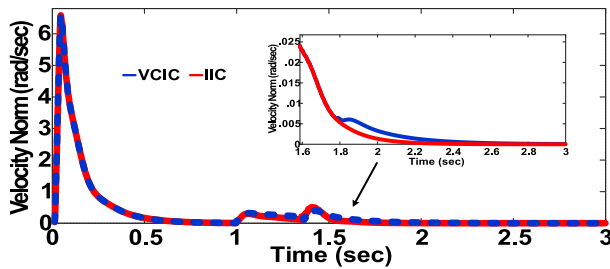


Fig. 20. Velocity vector norm (VCIC1 vs. IIC1).

cases, the robot is instructed to move toward a desired Cartesian position, via ramping functions, at which it collides with a surface. Fig. 17 illustrates the referential position ramp, in addition to the IIC1 and VCIC1 controller responses, while Fig. 18 displays the resulting impulse (force) values. Moreover, Fig. 19 exhibits the active impedance modulations and Fig. 20 demonstrates that during both simulations, the IIC and VCIC preimpact velocity norm magnitudes are identical.

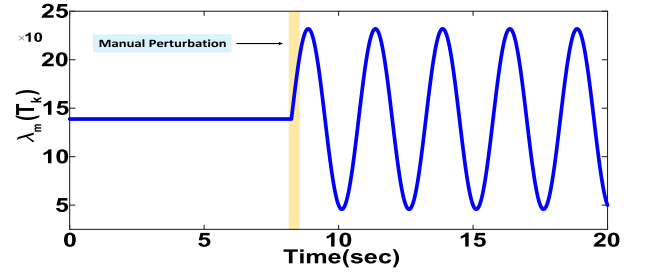
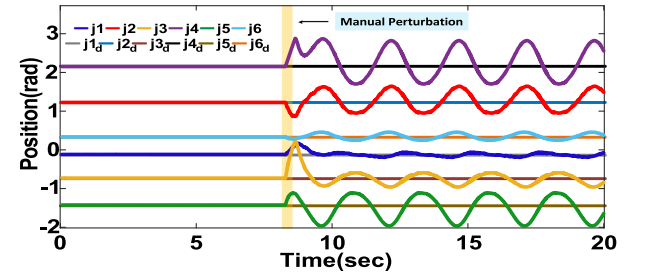
Fig. 21. Evolution of $\lambda_m(\mathbf{T}_K)$ value during VIC.

Fig. 22. Link position evolution before and after manual perturbation.

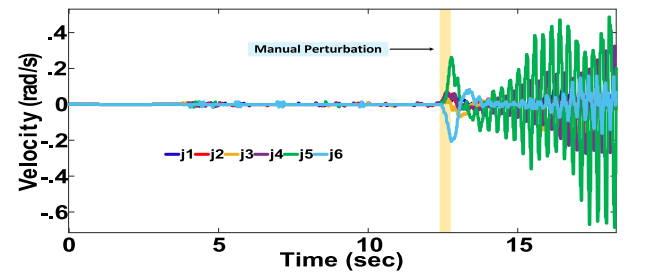


Fig. 23. Link velocities before and after a manual perturbation.

F. Practical Utility of the Stability Criteria

The Sawyer robot is initially at rest, with gains that ensure $\lambda_m(\mathbf{T}_K) > \alpha$ and $\lambda_m(\Xi_D) > 0$. Manual perturbation of the system triggers VIC at a gain modulation rate of 0.4 Hz (see Fig. 21) that periodically produces $\lambda_m(\mathbf{T}_K) < \alpha$, leading to joint oscillations around the desired equilibria (see Fig. 22). A different experiment entails the usage of gains yielding $\lambda_m(\Xi_D) = -2.37$, which render the system unstable when perturbed (see Fig. 23).

VI. CONCLUSION

The work described here introduces a novel control methodology, namely PPC, which is designed to suppress the adverse effects arising during VIC. The PPC technique enables the implementation of stable active VIC (gain-scheduling) methods on under-actuated, flexible-joint robots, with both linear and nonlinear passive stiffness elements. Since the PPC controller employs full-state feedback, one can utilize well-established

methods for systematic feedback gain generation, such as the LQR approach described herein.

Joint and Cartesian space versions of the PPC controller are introduced to permit intuitive definition of interaction tasks. Experiments involving VCIC and IIC, using both PPC and non-PPC schemes, enabled a comparative analysis involving the distinct methodologies. Experimental and simulation data reveals that employing VIC leads to the reduction of contact force magnitudes, as compared to using IIC.

APPENDIX A

Acquiring the steady-state form of the system comprising (1), (2), and (12), by setting $\dot{\mathbf{q}}_F = \ddot{\mathbf{q}}_F = 0$, and assuming that the robot is operating in “free motion” (i.e., $\tau_{ex} = 0$), leads to

$$\mathbf{T}_K \begin{bmatrix} \mathbf{q}_d - \mathbf{q} \\ \boldsymbol{\theta}_d - \boldsymbol{\theta} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\tau}_g(\mathbf{q}) - \boldsymbol{\tau}_{gd} \\ 0 \end{bmatrix}. \quad (40)$$

Applying a contraction mapping theorem based on $\mathbf{x} = [\mathbf{q}^T \boldsymbol{\theta}^T]^T$ and $\mathbf{H}(\mathbf{x}) = \mathbf{T}_K^{-1} \boldsymbol{\tau}(\mathbf{x}) - \mathbf{x}_d$, where $\boldsymbol{\tau}(\mathbf{x}) = [\boldsymbol{\tau}_g(\mathbf{q})^T - \boldsymbol{\tau}_{gd}^T \ 0]^T$, in order to acquire the inequality

$$\|\mathbf{H}(\mathbf{x}) - \mathbf{H}(\mathbf{y})\| \leq \lambda_M(\mathbf{T}_K^{-1}) \alpha \|\mathbf{x} - \mathbf{y}\| \quad (41)$$

guarantees uniqueness of (40), when $\lambda_m(\mathbf{T}_K) > \alpha$. Computing (15)’s gradient reveals its possession of a unique minimum around $\mathbf{q}_d$ and $\boldsymbol{\theta}_d$ in the following manner:

$$\nabla W = \mathbf{T}_{K0} \begin{bmatrix} \mathbf{q}_e \\ 0 \end{bmatrix} + \left[\boldsymbol{\tau}(\mathbf{x})^T \frac{(\boldsymbol{\theta}_E^T)^2}{2} \frac{\mathbf{q}_E^T \cdot \text{diag}(\boldsymbol{\theta}_E)}{2} \right]^T = 0 \quad (42)$$

with $\mathbf{T}_{K0} = \text{diag}([\mathbf{T}_K \ 0])$. Attaining its Hessian yields

$$\nabla^2 W = \mathbf{T}_{K0} - \begin{bmatrix} -\tau_\partial & 0 & 0 & \boldsymbol{\theta}_E/2 \\ 0 & 0 & \boldsymbol{\theta}_E & \mathbf{q}_E/2 \\ 0 & \boldsymbol{\theta}_E^T & 0 & 0 \\ \boldsymbol{\theta}_E^T/2 & \mathbf{q}_E^T/2 & 0 & 0 \end{bmatrix} > 0 \quad (43)$$

which verifies the latter’s positive definiteness, given that $\alpha \gg \boldsymbol{\theta}_E, \mathbf{q}_E$, thus confirming uniqueness of $(\mathbf{q}_E, \boldsymbol{\theta}_E)$.

APPENDIX B

The Lyapunov function’s (16) time derivative is

$$\begin{aligned} \dot{V} = & \dot{\mathbf{q}}_F^T \left(\begin{aligned} & \text{diag} \left(\begin{bmatrix} -\mathbf{C} & 0 \end{bmatrix} \right) \dot{\mathbf{q}}_F + \frac{\dot{\mathbf{M}}_T(\mathbf{q}) \dot{\mathbf{q}}_F}{2} - \boldsymbol{\tau}(\mathbf{x}) \\ & - \mathbf{T}_K \mathbf{q}_e - \boldsymbol{\Psi}_K \dot{\mathbf{q}}_F \\ & - \left[\frac{\dot{\boldsymbol{\Theta}}_{PI} I_d \left(\mathbf{q}_e^T \dot{\mathbf{T}}_K \mathbf{q}_e + \dot{\mathbf{q}}_F^T (\mathbf{T}_K^T - \mathbf{T}_K) \mathbf{q}_e \right)}{2} \right] \\ & + \frac{\dot{\mathbf{q}}_F^T \mathbf{T}_K \mathbf{q}_e}{2} + \frac{\mathbf{q}_e^T \mathbf{T}_K \dot{\mathbf{q}}_F}{2} + \dot{\mathbf{q}}_F^T \boldsymbol{\tau}(\mathbf{x}) \end{aligned} \right) \end{aligned}$$

$$+ \frac{\mathbf{q}_e^T \dot{\mathbf{T}}_K \mathbf{q}_e}{2} \quad (44)$$

which when considering that $\dot{\mathbf{q}}^T(\dot{\mathbf{M}} - 2\mathbf{C})\dot{\mathbf{q}} = 0$ and eliminating the gain-related terms yields (17), i.e., $\dot{V} = -\dot{\mathbf{q}}_F^T \boldsymbol{\Psi}_K \dot{\mathbf{q}}_F$. The $\boldsymbol{\Psi}_K$ matrix is positive definite if

$$\frac{1}{2} \begin{bmatrix} 2\mathbf{N} & \mathbf{K}_{D_{Jt}}^T \\ \mathbf{K}_{D_{Jt}} & 2(\mathbf{D} + \mathbf{K}_{D_{Mt}}) \end{bmatrix} > 0 \quad (45)$$

whose Schur complement yields (19). Since V and $\dot{\mathbf{q}}_F$ are bounded, while $\dot{V} \leq 0$, applying Barbalat’s lemma after differentiating $\dot{V}$, produces (20). Moreover, $m_l \leq \|\mathbf{M}_T(\mathbf{q})\| \leq m_u$, $\|\mathbf{C}\| \leq c_u \|\dot{\mathbf{q}}\|$, $\|\mathbf{D}\| \leq d_u$, $\|\mathbf{K}_T\| \leq k_u$, and $\|\boldsymbol{\tau}_g(\mathbf{q})\| \leq \tau_u$, where m_l , m_u , c_u , d_u , k_u , and τ_u denote positive constants, while $\mathbf{K}_T \in \mathbb{R}^{2n \times 2n}$ symbolizes the stiffness matrix. Therefore, $\ddot{\mathbf{q}}_F$ is bounded since

$$\ddot{\mathbf{q}}_F = \mathbf{M}_T^{-1} \left(\begin{bmatrix} 0 \\ \boldsymbol{\tau}_m \end{bmatrix} - \begin{bmatrix} \mathbf{C} + \mathbf{N} & 0 \\ 0 & \mathbf{D} \end{bmatrix} \dot{\mathbf{q}}_F - \mathbf{K}_T \mathbf{q}_F - \begin{bmatrix} \boldsymbol{\tau}_g(\mathbf{q}) \\ 0 \end{bmatrix} \right) \quad (46)$$

which comprises bounded terms. $\boldsymbol{\tau}_m$ is also bounded, since $\|\mathbf{q}_d\| \leq q_r$, with q_r signifying a positive constant. It is then concluded that $\dot{V}$ is uniformly continuous and $\lim_{t \rightarrow \infty} \dot{V}(\dot{\mathbf{q}}_F, t) = 0$. Substituting $\dot{\mathbf{q}}_F = 0$ into the closed-loop system, yields

$$\mathbf{K}(\mathbf{q} - \boldsymbol{\theta}) + \boldsymbol{\tau}_g(\mathbf{q}) = 0 \quad (47)$$

$$\mathbf{K}(\boldsymbol{\theta} - \mathbf{q}) = \mathbf{K}_{P_{Mt}} \boldsymbol{\theta}_E + \mathbf{K}_{P_{Jt}} \mathbf{q}_E + \boldsymbol{\tau}_{gd} \quad (48)$$

implying GAS at $\mathbf{q}_{eq}$ [50]. Alternatively, (46) is of the form $\ddot{\mathbf{x}}_F = f(\mathbf{x}_F, t)$, where $\mathbf{x}_F = [\dot{\mathbf{q}}_F^T \mathbf{q}_F^T]^T$. Assuming that

$$\dot{V} = -\dot{\mathbf{q}}_F^T \boldsymbol{\Psi}_K \dot{\mathbf{q}}_F \leq W_f(\dot{\mathbf{q}}_F) \leq 0 \quad (49)$$

where $W_f = -\dot{\mathbf{q}}_F^T \boldsymbol{\Omega} \dot{\mathbf{q}}_F$, with $0 < \boldsymbol{\Omega} < \boldsymbol{\Psi}_K$, it ought to be ensured that $\dot{V} \equiv 0$ (identically zero), as stipulated by the new invariance principle (NIP) [51]. Differentiating W_f yields

$$\dot{W}_f = -\ddot{\mathbf{q}}_F^T \boldsymbol{\Omega}_T \dot{\mathbf{q}}_F - \dot{\mathbf{q}}_F^T \dot{\boldsymbol{\Omega}} \dot{\mathbf{q}}_F \quad (50)$$

where $\boldsymbol{\Omega}_T = \boldsymbol{\Omega} + \boldsymbol{\Omega}^T$, which again indicates that $\ddot{V} = 0$ at $\dot{\mathbf{q}}_F = 0$, thus $\ddot{V} \equiv 0$. The second derivative of W_f produces

$$\begin{aligned} \ddot{W}_f = & -\ddot{\mathbf{q}}_F^T \boldsymbol{\Omega}_T \ddot{\mathbf{q}}_F - \ddot{\mathbf{q}}_F^T \boldsymbol{\Omega}_T \dot{\mathbf{q}}_F - \dot{\mathbf{q}}_F^T \ddot{\boldsymbol{\Omega}}_T \dot{\mathbf{q}}_F \\ & - \ddot{\mathbf{q}}_F^T \dot{\boldsymbol{\Omega}} \dot{\mathbf{q}}_F - \dot{\mathbf{q}}_F^T \dot{\boldsymbol{\Omega}} \ddot{\mathbf{q}}_F - \dot{\mathbf{q}}_F^T \ddot{\boldsymbol{\Omega}} \dot{\mathbf{q}}_F \end{aligned} \quad (51)$$

which is only zero at $\dot{\mathbf{q}}_F = \ddot{\mathbf{q}}_F = 0$. Substituting these vectors into the closed-loop system, yields (47)–(48), which possesses a unique solution at the origin $(\mathbf{q}_{eq})$. Since $\dot{V} \equiv 0$ and $f(\mathbf{x}_F, t)$ is bounded, Assumption 1 in [51] is satisfied, which according to the NIP proves GAS of this nonautonomous system. For uniformly locally Lipschitz continuous systems, [52, Th. 1] could be used instead.

APPENDIX C

Assuming that $\mathbf{J}_t$ is operated away from singularities and $\dot{\mathbf{q}}^T(\dot{\mathbf{M}} - 2\mathbf{C})\dot{\mathbf{q}} = 0$, a series of calculations produces

$$\dot{\mathbf{V}} = -\dot{\mathbf{x}}^T \mathbf{D}_{ct} \dot{\mathbf{x}} - \dot{\mathbf{q}}_F^T \begin{bmatrix} \mathbf{N} & \mathbf{0} \\ \mathbf{K}_{D_{jt}} & \mathbf{D} \end{bmatrix} \dot{\mathbf{q}}_F. \quad (52)$$

GAS can be proven by applying Barbalat's lemma and the NIP. The Cartesian controller's passivity is proven via

$$\dot{Q} = \dot{\mathbf{q}}^T \boldsymbol{\tau}_{ex} - \dot{\mathbf{q}}_F^T \boldsymbol{\Psi}_{K_C} \dot{\mathbf{q}}_F \leq \dot{\mathbf{q}}^T \boldsymbol{\tau}_{ex} \quad (53)$$

whose validity is ensured by positive definiteness of $\boldsymbol{\Psi}_{K_C}$, which, in turn, requires satisfaction of condition (19).

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